Bench	Task	Trials	Raw	+Workflow	+Skills	Raw Tok	WF Tok	SK Tok	ΔP_{SK-Raw}	ρ_{SK-Raw}
Terminal-Bench-2										
TB2	model-extraction-relu-logits	5	60.0%	0.0%	0.0%	482K	97K	468K	-60.0%	-2.8%
TB2	torch-tensor-parallelism	5	40.0%	0.0%	0.0%	174K	168K	153K	-40.0%	-12.4%
TB2	mcmc-sampling-stan	5	80.0%	0.0%	60.0%	10.8M	1.5M	10.5M	-20.0%	-2.6%
TB2	overfull-hbox	5	80.0%	40.0%	60.0%	555K	397K	550K	-20.0%	-0.9%
TB2	regex-chess	5	80.0%	100.0%	60.0%	3.0M	697K	900K	-20.0%	-70.4%
TB2	chess-best-move	5	80.0%	100.0%	80.0%	361K	144K	504K	+0.0%	+39.6%
TB2	path-tracing-reverse	5	80.0%	0.0%	80.0%	4.0M	916K	2.0M	+0.0%	-51.2%
TB2	query-optimize	5	40.0%	0.0%	40.0%	642K	275K	956K	+0.0%	+48.9%
TB2	tune-mjcf	5	20.0%	0.0%	20.0%	453K	219K	104K	+0.0%	-77.0%
TB2	build-pov-ray	5	80.0%	80.0%	100.0%	1.4M	1.5M	1.1M	+20.0%	-19.6%
TB2	crack-7z-hash	5	80.0%	0.0%	100.0%	986K	512K	2.0M	+20.0%	+103.4%
TB2	fix-code-vulnerability	5	80.0%	80.0%	100.0%	1.4M	941K	1.1M	+20.0%	-25.0%
TB2	large-scale-text-editing	5	80.0%	60.0%	100.0%	403K	284K	335K	+20.0%	-17.0%
TB2	log-summary-date-ranges	5	80.0%	100.0%	100.0%	163K	143K	131K	+20.0%	-19.8%
TB2	path-tracing	5	80.0%	0.0%	100.0%	2.2M	291K	1.7M	+20.0%	-21.7%
TB2	portfolio-optimization	5	80.0%	100.0%	100.0%	391K	289K	361K	+20.0%	-7.7%
TB2	rstan-to-pystan	5	80.0%	0.0%	100.0%	4.1M	1.0M	1.9M	+20.0%	-52.4%
TB2	sanitize-git-repo	5	80.0%	100.0%	100.0%	2.2M	1.1M	1.1M	+20.0%	-50.6%
TB2	winning-avg-corewars	5	80.0%	80.0%	100.0%	4.7M	573K	1.1M	+20.0%	-76.3%
TB2	qemu-startup	5	20.0%	100.0%	40.0%	758K	344K	1.8M	+20.0%	+130.9%
TB2	hf-model-inference	5	60.0%	80.0%	80.0%	1.1M	554K	346K	+20.0%	-69.1%
TB2	adaptive-rejection-sampler	5	20.0%	60.0%	60.0%	1.7M	651K	2.0M	+40.0%	+19.3%
TB2	dna-insert	5	20.0%	60.0%	60.0%	454K	395K	353K	+40.0%	-22.3%
TB2	financial-document-processor	5	60.0%	0.0%	100.0%	1.1M	423K	2.1M	+40.0%	+89.5%
TB2	mteb-leaderboard	5	60.0%	0.0%	100.0%	19.8M	1.7M	4.1M	+40.0%	-79.2%
TB2	pytorch-model-recovery	5	40.0%	40.0%	80.0%	794K	208K	360K	+40.0%	-54.7%
TB2	write-compressor	5	60.0%	0.0%	100.0%	248K	63K	401K	+40.0%	+61.3%
TB2	break-filter-js-from-html	5	40.0%	20.0%	100.0%	376K	202K	213K	+60.0%	-43.4%
TB2	cancel-async-tasks	5	40.0%	40.0%	100.0%	69K	96K	147K	+60.0%	+112.3%
TB2	extract-elf	5	40.0%	20.0%	100.0%	722K	245K	258K	+60.0%	-64.3%
TB2	pypi-server	5	20.0%	40.0%	100.0%	499K	364K	269K	+80.0%	-46.1%
SkillsBench										
SB	weighted-gdp-calc	5	80.0%	80.0%	20.0%	633K	478K	469K	-60.0%	-25.9%
SB	energy-market-pricing	5	60.0%	60.0%	0.0%	866K	1.0M	1.2M	-60.0%	+38.3%
SB	data-to-d3	5	60.0%	60.0%	20.0%	882K	634K	617K	-40.0%	-30.0%
SB	invoice-fraud-detection	5	60.0%	20.0%	20.0%	359K	273K	291K	-40.0%	-19.0%
SB	protein-expression-analysis	5	80.0%	100.0%	60.0%	529K	345K	436K	-20.0%	-17.6%
SB	grid-dispatch-operator	5	40.0%	20.0%	20.0%	635K	915K	381K	-20.0%	-39.9%
SB	paper-anonymizer	5	40.0%	0.0%	20.0%	940K	454K	608K	-20.0%	-35.3%
SB	trend-anomaly-causal-inference	5	20.0%	40.0%	0.0%	1.9M	4.8M	1.3M	-20.0%	-32.3%
SB	r2r-mpc-control	5	60.0%	60.0%	60.0%	526K	689K	514K	+0.0%	-2.2%
SB	dapt-intrusion-detection	5	80.0%	100.0%	100.0%	277K	576K	340K	+20.0%	+22.6%
SB	earthquake-phase-association	5	80.0%	60.0%	100.0%	2.1M	1.9M	2.7M	+20.0%	+25.6%
SB	glm-lake-mendota	5	80.0%	100.0%	100.0%	2.0M	1.5M	1.1M	+20.0%	-42.9%
SB	hvac-control	5	80.0%	80.0%	100.0%	220K	463K	324K	+20.0%	+47.5%
SB	adaptive-cruise-control	5	20.0%	20.0%	40.0%	5.7M	3.9M	2.8M	+20.0%	-50.8%
SB	fix-druid-loophole-cve	5	20.0%	40.0%	40.0%	13.4M	6.4M	9.2M	+20.0%	-31.4%
SB	fix-erlang-ssh-cve	5	60.0%	80.0%	80.0%	2.3M	1.6M	1.5M	+20.0%	-32.6%
SB	fix-visual-stability	5	60.0%	100.0%	80.0%	4.9M	4.4M	2.9M	+20.0%	-40.5%
SB	organize-messy-files	5	20.0%	0.0%	60.0%	602K	492K	639K	+40.0%	+6.1%
SB	video-filler-word-remover	5	40.0%	80.0%	80.0%	2.8M	1.7M	677K	+40.0%	-75.6%
SB	dialogue-parser	5	40.0%	80.0%	100.0%	488K	597K	406K	+60.0%	-16.7%
SB	financial-modeling-qa	5	40.0%	80.0%	100.0%	827K	627K	384K	+60.0%	-53.6%
SB	react-performance-debugging	5	40.0%	100.0%	100.0%	3.8M	2.7M	5.0M	+60.0%	+29.9%
SB	virtualhome-agent-planning	5	40.0%	100.0%	100.0%	284K	186K	325K	+60.0%	+14.1%
SB	gravitational-wave-detection	5	20.0%	80.0%	100.0%	2.5M	3.1M	545K	+80.0%	-78.6%
Terminal-Bench-Pro										
TB-Pro	fix-gomoku-win-detection	5	80.0%	100.0%	0.0%	92K	85K	142K	-80.0%	+54.3%
TB-Pro	implement-nonogram-puzzle-solver	5	60.0%	100.0%	20.0%	130K	333K	264K	-40.0%	+102.9%
TB-Pro	decode-go-ctf-credentials	5	80.0%	60.0%	60.0%	958K	968K	557K	-20.0%	-41.8%
TB-Pro	build-uci-chess-engine-to-beat-multiple-bots	5	40.0%	20.0%	20.0%	1.1M	743K	579K	-20.0%	-47.7%
TB-Pro	repair-broken-shell-data-pipeline	5	60.0%	100.0%	40.0%	759K	826K	515K	-20.0%	-32.2%
TB-Pro	flood-escape-path-planner	5	60.0%	80.0%	60.0%	214K	129K	328K	+0.0%	+53.5%
TB-Pro	improve-code-similarity-feature-extraction	5	80.0%	60.0%	80.0%	803K	972K	638K	+0.0%	-20.5%
TB-Pro	optimize-urdf-robot-for-pybullet	5	60.0%	80.0%	60.0%	1.6M	2.1M	1.1M	+0.0%	-32.6%
TB-Pro	portfolio-risk-analyzer-script	5	20.0%	0.0%	20.0%	182K	154K	298K	+0.0%	+63.3%
TB-Pro	regex-bitcoin-p2pkh-extraction	5	40.0%	100.0%	40.0%	65K	59K	129K	+0.0%	+98.7%
TB-Pro	reverse-engineer-kvstore-binary-protocol	5	80.0%	40.0%	80.0%	810K	294K	1.4M	+0.0%	+76.5%
TB-Pro	validate-and-solve-sudoku	5	60.0%	100.0%	60.0%	438K	312K	273K	+0.0%	-37.8%
TB-Pro	fix-xor-neural-network-instability	5	80.0%	100.0%	100.0%	544K	696K	862K	+20.0%	+58.4%
TB-Pro	secure-postgres-schema-migration	5	80.0%	100.0%	100.0%	634K	498K	420K	+20.0%	-33.8%
TB-Pro	train-fraud-detection-model	5	40.0%	40.0%	60.0%	180K	176K	148K	+20.0%	-18.0%
TB-Pro	train-loan-default-logreg-model	5	80.0%	100.0%	100.0%	147K	101K	158K	+20.0%	+7.6%
TB-Pro	implement-chemical-equilibrium-solver	5	20.0%	20.0%	40.0%	215K	430K	487K	+20.0%	+126.1%
TB-Pro	advanced-json-to-rfc4180-csv-converter	5	60.0%	80.0%	80.0%	401K	563K	747K	+20.0%	+86.5%
TB-Pro	present-known-plaintext-key-recovery	5	60.0%	100.0%	80.0%	5.9M	1.4M	1.7M	+20.0%	-70.6%
TB-Pro	implement-depgraph-dependency-resolver	5	20.0%	60.0%	60.0%	468K	252K	390K	+40.0%	-16.7%
TB-Pro	recover-and-sanitize-git-db-credential	5	20.0%	20.0%	60.0%	548K	371K	479K	+40.0%	-12.7%
TB-Pro	compare-cmake-build-types-performance	5	60.0%	100.0%	100.0%	111K	90K	120K	+40.0%	+8.5%
TB-Pro	mongodb-sales-aggregation-engine	5	60.0%	60.0%	100.0%	167K	131K	116K	+40.0%	-30.7%
TB-Pro	python-jacobi-eigen-solver-conversion	5	60.0%	60.0%	100.0%	198K	212K	208K	+40.0%	+5.3%
TB-Pro	setup-mlflow-sqlite-server	5	60.0%	80.0%	100.0%	1.0M	1.0M	204K	+40.0%	-80.2%
TB-Pro	text-image-ocr-pipeline	5	60.0%	100.0%	100.0%	109K	117K	266K	+40.0%	+144.4%
TB-Pro	train-matrix-factorization-embeddings	5	60.0%	20.0%	100.0%	351K	159K	179K	+40.0%	-49.1%
TB-Pro	train-sarsa-taxi-agent	5	60.0%	80.0%	100.0%	422K	723K	284K	+40.0%	-32.5%
TB-Pro	implement-tensor-parallel-matmul	5	40.0%	100.0%	100.0%	165K	299K	736K	+60.0%	+346.9%
TB-Pro	jq-github-contributor-report	5	40.0%	60.0%	100.0%	111K	117K	211K	+60.0%	+91.2%
TB-Pro	normalize-invoice-pdfs-to-csv	5	40.0%	0.0%	100.0%	212K	141K	237K	+60.0%	+11.7%
TB-Pro	python-sql-injection-sanitizer	5	40.0%	80.0%	100.0%	201K	140K	269K	+60.0%	+33.8%
TB-Pro	sokoban-bfs-move-with-fallback	5	20.0%	100.0%	100.0%	111K	164K	206K	+80.0%	+85.5%

bottomrule